

Comparison of Contraction Conditions

Banach (BC)

$$d(Tx, Ty) \leq k \cdot d(x, y)$$

Contraction constant $k < 1$

T is continuous

(Uniform convergence)

Chatterjea (CHC)

$$d(Tx, Ty) \leq r[d(Tx, y) + d(Ty, x)]$$

Coefficient $r \in [0, \frac{1}{2})$

Symmetric form of Kannan

(Independent from BC)

Kannan (KC)

$$d(Tx, Ty) \leq r[d(Tx, x) + d(Ty, y)]$$

Coefficient $r \in [0, \frac{1}{2})$

T is NOT necessarily continuous

(Local convergence)

Rhoades (1977)

*BC, KC, and CHC are
INDEPENDENT conditions*

No one implies another

**But all guarantee unique fixed points
in complete metric spaces**